THE STACK OF ETALE SHEAVES ON SCHEMES OVER A BASE FOR THE H-TOPOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the verification of the three conditions of the h-stack criterion, namely Zariski glueing of sheaves, descent along an fppf cover or a proper surjective cover, and the modification condition.

Editorial note. The label lemma-h-descent-etale-sheaves names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source etale-cohomology.tex –label lemma-h-descent-etale-sheaves –start 23256 –end 23285 –local-id 3574 returned RESULT=OK (L3574 free, occupancy FREE, public interval clear). Candidate taken from operations/pool-cards/stacks-etale-cohomology-halfB.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Criterion for a category fibred in groupoids to be an h-stack. More on Flatness, the lemma listing the three conditions (Zariski glueing, descent along surjective flat proper finitely presented morphisms, and the modification condition) that imply that a category fibred in groupoids over the h-site is a stack. Supplies the three conditions checked in the proof.
- Glueing of sheaves and descent for an fppf and for a proper surjective cover. Sites, the lemma on glueing of sheaves; Same chapter, the lemma giving descent of etale sheaves along an fppf cover and the lemma giving descent along a proper surjective cover. Supply condition (1) and condition (2) respectively.
- Descent of etale sheaves along a modification. Same chapter, the lemma establishing descent for the modification condition. Supplies condition (3).

Source reference. Original lemma 0EX9: sources/stacks-chow/tags.txt, source lines 18649–18649. Prerequisite pointer invoked at proof line(s) 23268; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GF2: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 23100–23100. Prerequisite pointer invoked at proof

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line(s) 23279; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GF4: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 23167–23167. Prerequisite pointer invoked at proof line(s) 23284; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GF0: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 23047–23047. Prerequisite pointer invoked at proof line(s) 23280; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04TR: sources/stacks-chow/tags.txt, source lines 6264–6264. Prerequisite pointer invoked at proof line(s) 23273; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Then the category fibred in groupoids*

$$p : \mathcal{S} \longrightarrow (Sch/S)_h$$

whose fibre category over U is the category $Sh(U_{étale})$ of sheaves on the small étale site of U is a stack in groupoids.

Proof. To prove the lemma we will check conditions (1), (2), and (3) of More on Flatness, Lemma 0EX9.

Condition (1) holds because we have glueing for sheaves (and Zariski coverings are étale coverings). See Sites, Lemma 04TR.

To see condition (2), suppose that $f : X \to Y$ is a surjective, flat, proper morphism of finite presentation over S with Y affine. Then we have descent for $\{X \to Y\}$ by either Lemma 0GF2 or Lemma 0GF0.

Condition (3) follows immediately from the more general Lemma 0GF4. □